Roll No.

Course: B.Tech Computer Engineering /
B.Tech COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)
Subject: Computer Organization and Architecture,
Subject Code: ETCS-203
Semester: III

Time: 03 Hours **Max Marks: 70**

Instructions to the Students:

1. This Question paper consists of two Sections. All sections are compulsory.
2. Section A comprises 10 questions of short answer type. All questions are compulsory. Each question carries 2 marks.
3. Section B comprises 8 long answer type questions out of which students must attempt any 5. Each question carries 10 marks.
4. Do not write anything on the question paper.

Q.No.	SECTION –A (SHORT ANSWER TYPE QUESTIONS)	Marks
1. a.	Define the concept of bus width and its impact on data transfer rates.	(2)
b.	What are the four main types of CPU architectures based on the organization of registers, and briefly explain each type?	(2)
c.	Differentiate between synchronous and asynchronous bus clocking mechanisms.	(2)
d.	Differentiate between fixed, variable, and hybrid instruction set formats, highlighting their respective advantages and disadvantages.	(2)
e.	Define data dependency in the context of pipelining. What are the solutions to handle data dependency issues?	(2)
f.	Define the term "branch target buffer" and explain its role in improving pipelining performance.	(2)
g.	Compare Serial and Parallel data transmission, highlighting their respective advantages and disadvantages.	(2)
h.	Explain Direct Memory Access (DMA) and its role in enhancing data transfer efficiency.	(2)
i.	Explain the format of microinstructions and discuss how they are stored and retrieved from Control Memory.	(2)
j.	Define the principle of Locality of Reference with examples.	(2)

SECTION –B (LONG ANSWER TYPE QUESTIONS)

2. Define RISC and CISC architectures. Discuss the key characteristics and design principles of RISC and CISC processors. Also compare and contrast them. (10)

3. Explain data representation methods with examples: (10)
   a) Signed Number Representation including sign and magnitude, two's complement, and one's complement.
   b) Fixed - point number representation.
   c) Floating - point number representation.
   d) Compare and contrast fixed-point and floating-point representations, highlighting their use cases and trade-offs.

4. Define addressing modes and its types. Explain the characteristics and applications of each addressing mode. Also provide examples of instructions utilizing each addressing mode. (10)

5. Explain the concept of pipelining, its types and how it affects throughput and speedup. Describe the organization and stages of an arithmetic pipeline with an example. (10)

6. Explain resource conflicts in a pipeline and discuss their resolution methods in detail with examples. (10)

7. List out and describe the mechanisms of asynchronous data transfer. Explain the significance of each in asynchronous communication with an example scenario. (10)

8. Define Cache Writing policies. Explain how each writing policy works and discuss their impact on system performance. Also analyze the trade-offs between these policies. (10)

9. Define Virtual Memory and what are the challenges and considerations in designing an efficient virtual memory system. Discuss the principles of page replacement in it using examples. (10)

===END OF PAPER===

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